

Han Gao

Undergraduate Researcher in Multimodal Artificial Intelligence

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Profile

Undergraduate student in Software Engineering at Xi'an Jiaotong University, conducting research on multimodal large language models, remote sensing foundation models, vision-language learning, and agentic AI.

Education

B.E. Xi'an Jiaotong University, Software Engineering
• Undergraduate student in the School of Electronics and Information Engineering. Xi'an, Shaanxi, China
Sept 2023 – present

Research Experience

Xi'an Jiaotong University, Undergraduate Researcher
Multimodal large language models for hyperspectral image change detection. Xi'an, Shaanxi, China
Jan 2026 – present

- Reframed pixel-level change detection as a vision-language question-answering task for semantic interpretation.
- Refined ground-truth labels and designed an automated pipeline for region-level captions and question-answer pairs.
- Adapted multimodal models with supervised fine-tuning and parameter-efficient fine-tuning, and conducted evaluation and analysis.

Xi'an Jiaotong University, Undergraduate Researcher
Structure-aware SAR-RGB cross-modal image retrieval. Xi'an, Shaanxi, China
Sept 2025 – Dec 2025

- Investigated CLIP-style alignment and contrastive learning across heterogeneous sensing modalities.
- Implemented strip-convolution modules to capture elongated ship structures.
- Completed model training, hyperparameter tuning, ablation studies, and comparative analysis.

Provincial Undergraduate Innovation Project, Core Team Member
Deep learning for cytological diagnosis of pleomorphic adenoma. Xi'an, Shaanxi, China
Sept 2025 – Apr 2026

- Contributed to literature review, technical planning, data preparation, model discussion, and experimental analysis.

Publications

HSICD: Multimodal Large Language Model for Hyperspectral Image Change Detection 2026

Han Gao, et al.
(IEEE Geoscience and Remote Sensing Magazine (under review))

A Unified Algorithmic Framework for Battery Life Prediction Based on CTMC and Monte Carlo Simulation 2026

Han Gao, et al.
(MEEML 2026)

Selected Projects

Multi-Agent Literature Research Assistant Mar 2026 – present

A tool-augmented multi-agent system for paper discovery, analysis, literature synthesis, and research knowledge management.

- Designed retrieval, analysis, and generation agents using MCP, A2A, and retrieval-augmented generation concepts.
- Integrated Zotero and Obsidian for research knowledge-base synchronization.
- Built agent tracing for observable coordination and tool-use workflows.

Skills

Programming: Python, C++, Java

Artificial Intelligence: PyTorch, Transformers, CLIP, LoRA, supervised fine-tuning, parameter-efficient fine-tuning

Research Areas: Multimodal large language models, remote sensing, vision-language learning, hyperspectral change detection, cross-modal retrieval, agentic AI